Embedding $\mathrm{SU}(2)$-covariant spin channels into $\mathrm{SU}(2)$-covariant quantum Markov semigroups

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Abstract. For a fixed irreducible spin representation, we express the known isotropic-spin generator classification and Wigner $6j$ rate transform in Hilbert–Schmidt normalized channel coordinates. A covariant channel with nontrivial sector spectrum $\boldsymbol{\eta}$ is a finite-time endpoint of a covariant quantum Markov semigroup if and only if every component is positive and $(M^{(j)})^{-1} \log \boldsymbol{\eta} \geq 0$. We derive the determinant magnitude of $M^{(j)}$, logarithmic endpoint inequalities through spin 2, and the exact volume of the embeddable region for every spin. For identity-normalized invertible differentiable covariant channel paths, the same cone characterizes CP-divisibility through the instantaneous logarithmic derivatives of the sector eigenvalues. The statement extends, with $\log |\boldsymbol{\eta}|$, to channel paths not normalized at the identity. Exact symbolic certificates accompany the low-spin matrices, polytopes, and volumes.

Keywords. SU(2)-covariant quantum channels; quantum Markov semigroups; GKSL generators; Markovian embedding; Wigner $6j$ symbols; CP-divisibility.

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1. Introduction

The embedding problem asks what a single quantum channel reveals about a possible continuous-time evolution: given Φ, does there exist a GKSL generator $\mathcal{L}$ and a time $t > 0$ such that $\Phi = e^{t\mathcal{L}}$? General finite-dimensional criteria must contend with matrix logarithm branches and conditional complete positivity [22, Eqs. (2)–(5)]. Symmetry changes that problem substantially when it forces both the channel and the generator to be scalar on a multiplicity-free decomposition.

For irreducible $\mathrm{SU}(2)$ representations, the static channel geometry is explicit. Al Nuwairan introduced a family called the EPOSIC channels and proved that they are precisely the extreme points of the covariant-channel simplex [1, Proposition 4.1]. The same vertices appear as Clebsch–Gordan channels in [4, Sec. 2], where low-rank entanglement and degradability questions are analyzed, in particular in Sec. 4. Cîrstoiu, Korzekwa, and Jennings describe the simplex in Liouville coordinates and derive the action of its extremal channels on irreducible tensor operators [6, Theorem 7 and Appendix B]. Aschieri, Ruba, and Solovej developed quantitative large-spin approximations and trace formulas for this simplex [2, Sec. 1]. Recent work on Weyl dynamical maps gives semigroup and non-Markovianity criteria in a different covariant family [23, Proposition 2]. In the same Weyl-covariant setting, Shahbeigi, Amaro-Alcalá, Puchała, and Życzkowski describe the log-convex geometry and volume of accessible channels [19, Corollary 4, Sec. IV, and Eq. (68)]. Rudnicki, Klimov, Muñoz, Leuchs, and Sánchez-Soto give a recent Choi–Kraus characterization of the static SU(2)-covariant channel family and its diagonal action on irreducible tensor sectors [18, Secs. 4–5, especially Eq. (38)]. At each simplex vertex, their Eq. (38) is the vertex eigenvalue formula used below; we record the exact conversion in Proposition 4.4.

The continuous-time single-spin problem also has direct antecedents. Verri and Gorini parametrized isotropic spin semigroups by interaction strengths and multipole relaxation rates [20, p. 1803]. Gorini, Verri, and Sudarshan subsequently corrected the complete-positivity condition and, in their single-spin specialization, gave the irreducible-rank generator form, mutually inverse $6j$-symbol transforms, and the resulting rate inequalities [9, Sec. 2, Eqs. (2.8), (2.15)–(2.16), and (2.20)–(2.21)]. Proposition 5.3 below translates their decay-rate inequalities into the present Hilbert–Schmidt and endpoint conventions.

More generally, Holevo's compact-group standard-form theorem shows that the completely positive part in a representation of a norm-continuous covariant semigroup may be chosen covariant and its Hamiltonian may be chosen to commute with the symmetry representation [12, Sec. 3.3.4, Theorem 3.3.5]; see also Holevo's original article [11, p. 211]. In the finite-dimensional irreducible spin setting, the multiplicity-free adjoint decomposition resolves this covariant completely positive part into one scalar rate per tensor rank. We give a self-contained derivation in the normalization needed for the endpoint calculation.

For time-dependent evolutions, the distinction between semigroup embeddability and divisibility is developed in [21, Secs. 2–4] and [7, Sec. 3]. The relation between CP-divisibility and time-local GKSL rates is treated in [17, especially Eqs. (1)–(2)], [10, Secs. II–III], and [15, Secs. 2–3]. These references use identity-normalized dynamical maps. Definition 7.1 also permits paths whose initial channel is not the identity; the distinction is stated explicitly there.

Here the symmetry is imposed on the continuous-time generator as well as on its endpoint. Thus "embeddable" below means embeddable by an SU(2)-covariant GKSL generator; no assertion is made about a noncovariant logarithm of the same endpoint. The argument uses three known inputs, which we restate and verify in the normalization of this paper:

1. the covariant channels form a simplex whose vertices act diagonally on irreducible tensor sectors;
2. every covariant generator is a unique nonnegative combination of tensor-rank dissipators; and
3. the multipole interaction strengths and relaxation rates are related by mutually inverse Wigner $6j$ transforms.

In these coordinates, the endpoint consequences are:

1. positivity of all sector eigenvalues together with $(M^{(j)})^{-1}\log\boldsymbol{\eta} \geq 0$ is necessary and sufficient for covariant endpoint embeddability;

2. the inverse-rate inequalities become explicit logarithmic-cone facets, listed through spin 2; and

3. along identity-normalized invertible differentiable covariant paths, the same cone applied to $d(\log\boldsymbol{\eta}(t))/dt$ characterizes CP-divisibility. For paths with a nonidentity initial channel, the corresponding statement uses $d(\log|\boldsymbol{\eta}(t)|)/dt$.

The additional calculations are the determinant-magnitude identity $|\det M^{(j)}| = 2j+1$, exact Euclidean volumes of the embeddable regions, and exact low-spin comparison polytopes. All displayed low-spin matrices and polytope intersections have symbolic certificates. Floating-point values are used only to render the spin-1 boundary curves and spin-3/2 boundary meshes whose defining equations are exact.

2. Tensor conventions and covariant channels

Let $j \in \frac{1}{2}\mathbb{N}_0$, let $\mathcal{H}_j$ be the spin-j irreducible unitary representation, and put $d = 2j+1$. Conjugation defines the unitary representation

$$\mathcal{U}_g(X) = U_j(g)XU_j(g)^\dagger \quad \text{on } B(\mathcal{H}_j).$$

Definition 2.1 (Dynamical conventions)**.** A channel on a finite-dimensional matrix algebra is a complex-linear, completely positive, trace-preserving map; all channels below are written in the Schrödinger picture. A linear map $A : B(\mathcal{H}_j) \to B(\mathcal{H}_j)$ is SU(2)-covariant when

$$A \circ \mathcal{U}_g = \mathcal{U}_g \circ A \qquad (g \in \mathrm{SU}(2)). \tag{2.1}$$

A norm-continuous quantum Markov semigroup is a family $\{e^{t\mathcal{L}}\}_{t\geq 0}$ of channels satisfying the semigroup law. Its generator $\mathcal{L}$ is covariant when it satisfies (2.1), equivalently when every $e^{t\mathcal{L}}$ is covariant.

The Clebsch–Gordan decomposition is multiplicity free:

$$B(\mathcal{H}_j) \cong \mathcal{H}_j \otimes \mathcal{H}_j^* \cong \bigoplus_{\ell=0}^{2j} V_\ell. \tag{2.2}$$

When $j = 0$, the algebra is $\mathbb{C}$, the only channel is the identity, and the only trace-preserving generator is zero. All matrix and inequality statements below then have empty index sets. We retain this case in the general statements and focus the nontrivial formulas on $j > 0$.

We use the Hilbert–Schmidt orthonormal spherical tensors of Breuer [3, Appendix A, Eqs. (A1)–(A6)]. In the angular momentum basis they are

$$\langle j, m_1 | T_q^{(k)} | j, m_2 \rangle = \sqrt{2k+1}\,(-1)^{j-m_1} \begin{pmatrix} j & j & k \\ m_1 & -m_2 & -q \end{pmatrix}. \tag{2.3}$$

Thus

$$\mathrm{Tr}\Big((T_q^{(k)})^\dagger T_{q'}^{(k')}\Big) \;=\; \delta_{kk'}\delta_{qq'}, \qquad\qquad T_0^{(0)} \;=\; I/\sqrt{d}, \tag{2.4}$$

$$\mathcal{U}_g(T_q^{(k)}) \;=\; \sum_{r=-k}^{k} D_{rq}^{(k)}(g)T_r^{(k)}, \qquad\qquad (T_q^{(k)})^\dagger \;=\; (-1)^q T_{-q}^{(k)}. \tag{2.5}$$

Lemma 2.2 *(Scalar tensor sum).* *For $0 \le k \le 2j$,*

$$\sum_{q=-k}^{k}(T_q^{(k)})^\dagger T_q^{(k)} \;=\; \sum_{q=-k}^{k} T_q^{(k)}(T_q^{(k)})^\dagger \;=\; \frac{2k+1}{d}I. \tag{2.6}$$

Proof. The first sum is invariant under conjugation by (2.5), hence it is scalar by irreducibility of U_j. Its trace is $2k+1$ by (2.4), which gives the first identity. The adjoint relation in (2.5) permutes the summands of the two sums, giving the second identity. ∎

Definition 2.3 (Covariant rank dissipators)**.** For $1 \le k \le 2j$, define

$$\mathcal{D}_k(X) \;=\; \sum_{q=-k}^{k}\Big(T_q^{(k)}X(T_q^{(k)})^\dagger - \frac{1}{2}\{(T_q^{(k)})^\dagger T_q^{(k)}, X\}\Big) \tag{2.7}$$

$$\;=\; \sum_{q=-k}^{k} T_q^{(k)}X(T_q^{(k)})^\dagger - \frac{2k+1}{d}X. \tag{2.8}$$

Lemma 2.4 *(Dirichlet form).* *For every $X \in B(\mathcal{H}_j)$ and $1 \le k \le 2j$,*

$$\langle X, \mathcal{D}_k(X)\rangle_{\mathrm{HS}} \;=\; -\frac{1}{2}\sum_{q=-k}^{k}\big\|[T_q^{(k)}, X]\big\|_{\mathrm{HS}}^2 \;\le\; 0. \tag{2.9}$$

Consequently every entry $M_{\ell k}^{(j)}$ introduced below is nonpositive, and every covariantly embeddable channel satisfies $0 < \eta_\ell \le 1$.

Proof. Write $A_q = T_q^{(k)}$ and $c = (2k+1)/d$. The set $\{A_q\}_{q=-k}^{k}$ is closed under adjoints up to signs, and Lemma 2.2 gives $\sum_q A_q^\dagger A_q = \sum_q A_q A_q^\dagger = cI$. Expanding the squared commutators therefore yields

$$\frac{1}{2}\sum_q \|[A_q, X]\|_{\mathrm{HS}}^2 \;=\; c\|X\|_{\mathrm{HS}}^2 - \mathrm{Re}\sum_q \mathrm{Tr}(X^\dagger A_q X A_q^\dagger).$$

The Kraus sum is Hilbert–Schmidt self-adjoint because $A_q^\dagger = (-1)^q A_{-q}$, so the trace sum on the right is real. Comparison with (2.8) proves (2.9). Taking $X = T_m^{(\ell)}$ gives $M_{\ell k}^{(j)} \le 0$. The final assertion follows from $\eta_\ell = \exp(t\sum_k M_{\ell k}^{(j)}\gamma_k)$. ∎

Proposition 2.5 *(Rank-one angular-momentum diffusion).* *Assume $j > 0$, and let J_x, J_y, J_z be the spin-j angular momentum operators. Then*

$$\mathcal{D}_1(X) \;=\; -\frac{3}{2j(j+1)(2j+1)}\sum_{a=x,y,z}[J_a,[J_a,X]]. \tag{2.10}$$

In particular, its eigenvalue on V_ℓ is

$$M^{(j)}_{\ell 1} \;=\; -\frac{3\ell(\ell+1)}{2j(j+1)(2j+1)}. \tag{2.11}$$

Proof. Rotation invariance and $\sum_a J_a^2 \;=\; j(j+1)I$ give

$$\mathrm{Tr}(J_a J_b) \;=\; s\delta_{ab}, \qquad s \;=\; \frac{j(j+1)(2j+1)}{3}.$$

With spherical components $J_0 \;=\; J_z$ and $J_{\pm 1} \;=\; \mp(J_x \pm iJ_y)/\sqrt{2}$, the convention (2.3) is $T^{(1)}_q \;=\; J_q/\sqrt{s}$. Changing unitarily from spherical to Cartesian components in (2.8) therefore gives

$$\mathcal{D}_1(X) \;=\; \frac{1}{s}\left(\sum_a J_a X J_a - j(j+1)X\right),$$

which is (2.10). The adjoint Casimir $\sum_a [J_a, [J_a, \cdot]]$ acts on the spin-ℓ summand V_ℓ by $\ell(\ell+1)$, proving (2.11). ∎

Proposition 2.6 *(Depolarizing direction)***.** *Let*

$$\mathcal{P}_0(X) \;=\; \frac{\mathrm{Tr}(X)}{d} I. \tag{2.12}$$

Then

$$\mathcal{P}_0 - \mathrm{id} \;=\; \frac{1}{d}\sum_{k=1}^{2j} \mathcal{D}_k. \tag{2.13}$$

Consequently the equal-rate point $\gamma_k \;=\; 1/d$ generates the standard depolarizing semigroup

$$e^{t(\mathcal{P}_0 - \mathrm{id})}(X) \;=\; e^{-t}X + (1-e^{-t})\frac{\mathrm{Tr}(X)}{d} I, \tag{2.14}$$

whose nontrivial sector eigenvalues are all e^{-t}.

Proof. The tensors $T^{(k)}_q$, over all k, q, form a Hilbert–Schmidt orthonormal basis of $B(\mathcal{H}_j)$. Completeness of any such matrix basis gives

$$\sum_{k=0}^{2j}\sum_{q=-k}^{k} T^{(k)}_q X (T^{(k)}_q)^\dagger \;=\; \mathrm{Tr}(X) I.$$

The $k \;=\; 0$ term is X/d, and $\sum_{k=1}^{2j}(2k+1) \;=\; d^2 - 1$. Summing (2.8) over $k \;\geq\; 1$ therefore gives $\mathrm{Tr}(X)I - dX$, which is (2.13). Since $\mathcal{P}_0$ is a projection and $\mathcal{P}_0\,\mathrm{id} \;=\; \mathrm{id}\,\mathcal{P}_0 \;=\; \mathcal{P}_0$, its exponential is (2.14). ∎

For later comparison with the complete channel simplex, set

$$\mathcal{E}_k(X) \;=\; \frac{d}{2k+1}\sum_{q=-k}^{k} T^{(k)}_q X (T^{(k)}_q)^\dagger, \qquad 0 \;\leq\; k \;\leq\; 2j. \tag{2.15}$$

Proposition 2.7 *(Covariant-channel simplex)*. *The maps $\mathcal{E}_0, \dots, \mathcal{E}_{2j}$ are precisely the vertices of the simplex of* SU(2)*-covariant channels on $B(\mathcal{H}_j)$. Moreover,*

$$\mathcal{D}_k \;=\; \frac{2k+1}{d}(\mathcal{E}_k - \mathrm{id}). \tag{2.16}$$

Proof. Lemma 2.2 shows that every $\mathcal{E}_k$ is trace preserving and unital. Under column vectorization, its Choi matrix is $d/(2k+1)$ times the orthogonal projector onto the rank-k summand of $\mathcal{H}_j \otimes \mathcal{H}_j^*$. The irreducible Choi projectors are exactly the simplex vertices by [2, Sec. 1, Eqs. (1.34)–(1.37), and Appendix A]. Equation (2.16) follows by comparing (2.8) and (2.15). ∎

3. Classification of covariant generators

Theorem 3.1 *(Covariant-generator decomposition)*. *A linear map $\mathcal{L} : B(\mathcal{H}_j) \to B(\mathcal{H}_j)$ generates a norm-continuous* SU(2)*-covariant quantum Markov semigroup if and only if there are unique numbers $\gamma_1, \dots, \gamma_{2j} \geq 0$ such that*

$$\mathcal{L} \;=\; \sum_{k=1}^{2j} \gamma_k \mathcal{D}_k. \tag{3.1}$$

In particular, its canonical traceless GKS Hamiltonian is zero, and the generator is self-adjoint for the Hilbert–Schmidt inner product.

The covariance statement behind this decomposition is the finite-dimensional compact-group case of Holevo's standard-form result [12, Sec. 3.3.4, Theorem 3.3.5]. The proof below identifies the unique tensor-rank coefficients in the convention (2.4). In the single-spin specialization, the resulting rank-block form is also the normalization-specific version of [9, Eqs. (2.8) and (2.11)].

Proof. Choose the tensors $T_q^{(k)}$, $1 \leq k \leq 2j$, as an orthonormal basis of the traceless operators. Although the spherical tensors need not be self-adjoint, they are related to any self-adjoint Hilbert–Schmidt orthonormal basis by a unitary basis change. Under that change the GKS matrix is conjugated unitarily, so its positivity and the uniqueness of the canonical traceless data are preserved. The canonical finite-dimensional Gorini–Kossakowski–Sudarshan–Lindblad representation, as stated in [8, Theorem 2.2 and Lemma 2.3] (see also [13, Theorem 2]), gives unique data $H = H^\dagger$, $\operatorname{Tr} H = 0$, and a positive semidefinite matrix C on the traceless operator space such that

$$\mathcal{L}(X) \;=\; -i[H, X] + \sum_{a,b} C_{ab} \left(F_a X F_b^\dagger - \frac{1}{2}\{F_b^\dagger F_a, X\} \right), \tag{3.2}$$

where $\{F_a\}$ is the chosen tensor basis.

For $g \in \mathrm{SU}(2)$, conjugating (3.2) by $\mathcal{U}_g$ replaces H by $U_j(g) H U_j(g)^\dagger$ and C by its conjugate under the representation of SU(2) on the traceless operator space. Covariance of $\mathcal{L}$, followed by uniqueness of the canonical data, therefore gives

$$U_j(g) H U_j(g)^\dagger \;=\; H, \qquad C\,\mathcal{U}_g \;=\; \mathcal{U}_g C \quad (g \in \mathrm{SU}(2)). \tag{3.3}$$

Schur's lemma and irreducibility of U_j make H scalar; the condition $\operatorname{Tr} H = 0$ gives $H = 0$.

By (2.2), the traceless operator space is the multiplicity-free sum $\bigoplus_{k=1}^{2j} V_k$. A second application of Schur's lemma gives

$$C = \bigoplus_{k=1}^{2j} \gamma_k I_{V_k}. \tag{3.4}$$

Positivity of C is equivalent to $\gamma_k \geq 0$ for every k. Substitution of (3.4) into (3.2) gives (3.1).

Conversely, each $\mathcal{D}_k$ is in GKSL form and is covariant by (2.5). Hence every nonnegative combination in (3.1) generates a covariant quantum Markov semigroup. Uniqueness of the canonical GKS matrix gives uniqueness of the coefficients. Finally, the adjoint relation in (2.5) shows that the Kraus sum in (2.8) is Hilbert–Schmidt self-adjoint. ∎

4. The exact $6j$-symbol matrix

Lemma 4.1 *(Sector diagonalization and reality).* *If $A : B(\mathcal{H}_j) \to B(\mathcal{H}_j)$ is covariant, then there are scalars $a_0, \ldots, a_{2j}$ such that*

$$A(T_m^{(\ell)}) = a_\ell T_m^{(\ell)} \quad (-\ell \leq m \leq \ell). \tag{4.1}$$

If A is Hermiticity preserving, every a_ℓ is real. If A is a channel, then $a_0 = 1$.

Proof. The irreducible summands in (2.2) are pairwise inequivalent and occur with multiplicity one, so (4.1) follows from Schur's lemma. Since $(T_0^{(\ell)})^\dagger = T_0^{(\ell)}$, Hermiticity preservation forces $a_\ell T_0^{(\ell)} = \overline{a_\ell} T_0^{(\ell)}$, hence $a_\ell \in \mathbb{R}$. Finally, covariance makes $A(I)$ invariant under U_j, so $A(I) = cI$; trace preservation gives $c = 1$. ∎

For a covariant generator, write $\mathcal{L}(T_m^{(\ell)}) = \lambda_\ell T_m^{(\ell)}$.

Lemma 4.2 *(Tensor recoupling).* *For $0 \leq k, \ell \leq 2j$,*

$$\sum_{q=-k}^{k} T_q^{(k)} T_m^{(\ell)} (T_q^{(k)})^\dagger = a_{\ell k} T_m^{(\ell)}, \tag{4.2}$$

where

$$a_{\ell k} = (2k+1)(-1)^{2j+k+\ell} \begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix}. \tag{4.3}$$

Proof. The left side of (4.2) defines a covariant map, so it is scalar on V_ℓ. We calculate the scalar with the invariant unitary

$$\beta : \mathcal{H}_j^* \longrightarrow \mathcal{H}_j, \qquad \beta\langle j, m| = (-1)^{j-m} |j, -m\rangle. \tag{4.4}$$

Fix column vectorization by

$$\operatorname{vec}(|j, m_1\rangle\langle j, m_2|) = |j, m_1\rangle \otimes \langle j, m_2|. \tag{4.5}$$

It satisfies $\mathrm{vec}(AXB^\dagger) = (A \otimes \overline{B})\,\mathrm{vec}(X)$. Hence the Kraus map in (4.2) is represented on $\mathcal{H}_j \otimes \mathcal{H}_j^*$ by $\sum_q T_q^{(k)} \otimes \overline{T_q^{(k)}}$.

In the angular-momentum basis and its dual, the matrix of β is the rotation-by-π matrix about the y-axis in [3, Eq. (3.10)], up to an overall phase which cancels under conjugation. Self-duality of the spin representation is the matrix identity

$$\beta\,\overline{U_j(g)}\,\beta^\dagger \;=\; U_j(g) \qquad (g \,\in\, \mathrm{SU}(2)), \tag{4.6}$$

so $I \otimes \beta$ intertwines $U_j \otimes \overline{U_j}$ with $U_j \otimes U_j$. In the chosen angular-momentum basis, the Clebsch–Gordan coefficients and hence the tensors are real, and [3, Eq. (A6)] therefore states $\beta(T_q^{(k)})^\mathsf{T}\beta^\dagger \;=\; (-1)^k T_q^{(k)}$. Taking adjoints gives

$$\beta\,\overline{T_q^{(k)}}\,\beta^\dagger \;=\; (-1)^k (T_q^{(k)})^\dagger. \tag{4.7}$$

Consequently, after applying $I \otimes \beta$, the transfer matrix of the Kraus map is $(-1)^k Q_k$, where

$$Q_k \;=\; \sum_{q=-k}^{k} T_q^{(k)} \otimes (T_q^{(k)})^\dagger$$

is exactly Breuer's invariant tensor [3, Eq. (2.7)]. Let P_ℓ be the projector onto the coupled spin-ℓ summand of $\mathcal{H}_j \otimes \mathcal{H}_j$. The unitary $I \otimes \beta$ is an SU(2)-intertwiner. By (2.5), it maps $\mathrm{span}\{\mathrm{vec}(T_m^{(\ell)}) \,:\, -\ell \,\le\, m \,\le\, \ell\}$ onto a spin-ℓ summand; multiplicity one identifies its image with $P_\ell(\mathcal{H}_j \otimes \mathcal{H}_j)$. Taking the trace on that $(2\ell+1)$-dimensional summand therefore gives

$$a_{\ell k} \;=\; (-1)^k \frac{\mathrm{Tr}(Q_k P_\ell)}{2\ell+1}. \tag{4.8}$$

Breuer's Eqs. (2.13)–(2.14), with $K \,=\, k$, $J \,=\, \ell$, and $j_1 \,=\, j_2 \,=\, j$, state

$$\mathrm{Tr}(Q_k P_\ell) \;=\; \sqrt{(2k+1)(2\ell+1)}\, L_{k\ell}, \tag{4.9}$$

$$L_{k\ell} \;=\; \sqrt{(2k+1)(2\ell+1)}(-1)^{2j+\ell} \begin{Bmatrix} j & j & \ell \\ j & j & k \end{Bmatrix}. \tag{4.10}$$

Substitution in (4.8), followed by the tetrahedral symmetry

$$\begin{Bmatrix} j & j & \ell \\ j & j & k \end{Bmatrix} \;=\; \begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix},$$

gives (4.3). ■

Corollary 4.3 *(Channel-vertex eigenvalues)***.** *The simplex vertex $\mathcal{E}_k$ has tensor-sector eigenvalues*

$$v_{\ell k}^{(j)} \;=\; (2j+1)(-1)^{2j+k+\ell} \begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix}, \qquad 0 \,\le\, k, \ell \,\le\, 2j. \tag{4.11}$$

In particular, $v_{0k}^{(j)} \;=\; v_{\ell 0}^{(j)} \;=\; 1$.

Proof. Multiply (4.3) by the normalization $d/(2k+1)$ in (2.15). The cases $k \,=\, 0$ and $\ell \,=\, 0$ also follow directly from $\mathcal{E}_0 \,=\, \mathrm{id}$ and trace preservation. ■

Proposition 4.4 *(Static normalization dictionary)***.** *Let α_k denote the coefficients used in [18, Eqs. (26)–(31)], normalized by*

$$\sum_{k=0}^{2j}(2k+1)\alpha_k \;=\; 1. \tag{4.12}$$

The barycentric weights in (4.24) are

$$p_k \;=\; (2k+1)\alpha_k. \tag{4.13}$$

With this substitution, their sector formula [18, Eq. (38)] becomes

$$\eta_\ell \;=\; \sum_{k=0}^{2j} p_k v_{\ell k}^{(j)}. \tag{4.14}$$

Proof. The rank-k Kraus operators in the cited normalization are $\sqrt{d\alpha_k}\,T_q^{(k)}$. A contribution $p_k\mathcal{E}_k$ in (2.15) has Kraus operators $\sqrt{p_k d/(2k+1)}\,T_q^{(k)}$, proving (4.13). Substitution in the cited sector formula and comparison with (4.11) gives (4.14). ∎

Theorem 4.5 *(Generator eigenvalue matrix and inverse)***.** *For $1 \le \ell, k \le 2j$, set*

$$M_{\ell k}^{(j)} \;=\; (2k+1)\left[(-1)^{2j+k+\ell}\begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix} - \frac{1}{2j+1}\right]. \tag{4.15}$$

If $\mathcal{L} \;=\; \sum_k \gamma_k\mathcal{D}_k$, then

$$(\lambda_1,\ldots,\lambda_{2j})^{\mathsf{T}} \;=\; M^{(j)}(\gamma_1,\ldots,\gamma_{2j})^{\mathsf{T}}. \tag{4.16}$$

The matrix is invertible, with

$$\big((M^{(j)})^{-1}\big)_{k\ell} \;=\; (2\ell+1)(-1)^{2j+k+\ell}\begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix}. \tag{4.17}$$

Moreover,

$$\left|\det M^{(j)}\right| \;=\; 2j+1. \tag{4.18}$$

Proof. Equations (2.8) and (4.3) give (4.15) and (4.16). It remains to invert the matrix.

Put $n = 2j$, $w_r = 2r+1$, and let $V = (v_{\ell k}^{(j)})_{0\le \ell,k\le n}$ be the full vertex eigenvalue matrix from Corollary 4.3. Let $W = \mathrm{diag}(w_0,\ldots,w_n)$ and $S = \mathrm{diag}((-1)^0,\ldots,(-1)^n)$. Breuer's recoupling matrix is

$$L_{k\ell} \;=\; \sqrt{w_k w_\ell}\,(-1)^{n+\ell}\begin{Bmatrix} j & j & \ell \\ j & j & k \end{Bmatrix};$$

the orthogonality relation used here is

$$\sum_{\ell=0}^{n} w_\ell \begin{Bmatrix} j & j & \ell \\ j & j & k \end{Bmatrix}\begin{Bmatrix} j & j & \ell \\ j & j & k' \end{Bmatrix} \;=\; \frac{\delta_{kk'}}{w_k}. \tag{4.19}$$

It is the $j_1 = j_2 = j$ specialization of the sum rule following [3, Eq. (2.14)]; see also [16, Sec. 34.5(iv)]. It says that L is orthogonal. The tetrahedral symmetry in [16, Sec. 34.5(ii)] and (4.11) give

$$V = (n+1)W^{-1/2}L^{\mathsf{T}}W^{-1/2}S, \qquad V^{-1} = \frac{1}{n+1}SW^{1/2}LW^{1/2}. \tag{4.20}$$

Hence

$$(V^{-1})_{k\ell} = \frac{w_k w_\ell}{n+1}(-1)^{n+k+\ell}\begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix}. \tag{4.21}$$

Since the first row and first column of V consist of ones, subtracting its first row from all other rows gives the block matrix

$$\begin{pmatrix} 1 & \mathbf{1}^{\mathsf{T}} \\ 0 & B \end{pmatrix}, \qquad B_{\ell k} = v_{\ell k}^{(j)} - 1 \quad (1 \le \ell, k \le n).$$

The lower-right block of V^{-1} is therefore B^{-1}. Moreover, (2.16) gives

$$M^{(j)} = B\,\mathrm{diag}(w_1/d, \dots, w_n/d).$$

Combining this identity with (4.21) proves (4.17).

Finally, (4.20) gives

$$|\det V| = \frac{d^d}{\prod_{r=0}^{n} w_r}.$$

Row subtraction shows $\det V = \det B$, and multiplying the columns of B by w_k/d, $1 \le k \le n$, gives $|\det M^{(j)}| = d$, proving (4.18). ■

Corollary 4.6 *(Dissipator–vertex conversion).* *The vertex and generator eigenmatrices satisfy*

$$M_{\ell k}^{(j)} = \frac{2k+1}{2j+1}(v_{\ell k}^{(j)} - 1). \tag{4.22}$$

Proof. Restrict (2.16) to V_ℓ. ■

Corollary 4.7 *(Complete-positivity halfspaces).* *Let A be a covariant, Hermiticity-preserving, trace-preserving map with sector eigenvalues $\eta_0 = 1, \eta_1, \dots, \eta_{2j}$. Define*

$$p_k = \frac{2k+1}{2j+1}\sum_{\ell=0}^{2j}(2\ell+1)(-1)^{2j+k+\ell}\begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix}\eta_\ell, \qquad 0 \le k \le 2j. \tag{4.23}$$

Then A is completely positive if and only if $p_k \ge 0$ for every k. In that case $\sum_k p_k = 1$ and

$$A = \sum_{k=0}^{2j} p_k \mathcal{E}_k. \tag{4.24}$$

Proof. In sector coordinates, $\boldsymbol{\eta} = V\boldsymbol{p}$, where V is the full vertex matrix used in (4.20). Formula (4.21) says that $\boldsymbol{p} = V^{-1}\boldsymbol{\eta}$ is exactly (4.23). Proposition 2.7 identifies complete positivity with nonnegative barycentric weights. The equality $\sum_k p_k = \eta_0 = 1$ follows from the first row of V. ■

5. Endpoint embeddability

Theorem 5.1 *(Covariant Markovian embedding criterion). Let Φ be an* SU(2)*-covariant channel on $B(\mathcal{H}_j)$, with tensor-sector eigenvalues $\eta_0 = 1, \eta_1, \ldots, \eta_{2j}$. There are $t > 0$ and an* SU(2)*-covariant GKSL generator $\mathcal{L}$ such that $\Phi = e^{t\mathcal{L}}$ if and only if*

$$\eta_\ell > 0 \quad (1 \le \ell \le 2j), \qquad (M^{(j)})^{-1} \begin{pmatrix} \log \eta_1 \\ \vdots \\ \log \eta_{2j} \end{pmatrix} \in \mathbb{R}^{2j}_{\ge 0}. \tag{5.1}$$

Proof. If $\Phi = e^{t\mathcal{L}}$, Theorem 3.1 gives $\mathcal{L} = \sum_k \gamma_k \mathcal{D}_k$ with $\gamma_k \ge 0$. Its sector eigenvalues are real by Theorem 4.5, hence $\eta_\ell = e^{t\lambda_\ell} > 0$, and

$$\log \boldsymbol{\eta} = M^{(j)}(t\boldsymbol{\gamma}).$$

This proves necessity. Conversely, define $r = (M^{(j)})^{-1} \log \boldsymbol{\eta} \ge 0$ and, taking $t = 1$, $\mathcal{L} = \sum_k r_k \mathcal{D}_k$. Theorem 3.1 makes $\mathcal{L}$ a covariant GKSL generator. The maps $e^{\mathcal{L}}$ and Φ have the same scalar on every summand in (2.2); therefore they are equal. ∎

Corollary 5.2 *(General logarithmic-cone facets and multiplicative geometry). For a covariant channel with positive sector eigenvalues, define*

$$r_k = \sum_{\ell=1}^{2j} (2\ell+1)(-1)^{2j+k+\ell} \begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix} \log \eta_\ell, \qquad 1 \le k \le 2j. \tag{5.2}$$

Then the channel is covariantly embeddable if and only if the $2j$ inequalities $r_k \ge 0$ hold. The vector r is the unique integrated-rate vector. In logarithmic eigenvalue coordinates, the facet $r_k = 0$ is the image of the generator face on which the rank-k dissipator is absent; its image in the $\boldsymbol{\eta}$-coordinates is generally curved.

The embeddable eigenvalue set is closed under componentwise multiplication and under weighted geometric interpolation: if $\boldsymbol{\eta}$ and $\boldsymbol{\zeta}$ are embeddable and $0 \le s \le 1$, then so are

$$\boldsymbol{\eta}\boldsymbol{\zeta} \quad \text{and} \quad \boldsymbol{\eta}^s \boldsymbol{\zeta}^{1-s}, \tag{5.3}$$

where all operations are componentwise.

Proof. Equation (5.2) is $(M^{(j)})^{-1} \log \boldsymbol{\eta}$ written using (4.17), so the first assertion is Theorem 5.1. In logarithmic coordinates, the two operations in (5.3) add the integrated rates and take their convex combinations, respectively. Both preserve $\mathbb{R}^{2j}_{\ge 0}$. Equivalently, all covariant generators commute because they are diagonal on the tensor-rank sectors, so the corresponding exponentials compose by addition of their rates. ∎

Proposition 5.3 *(Relaxation-rate normalization dictionary). Fix $t > 0$, and write the sector decay rates as*

$$\chi_\ell = -\frac{1}{t} \log \eta_\ell, \qquad 1 \le \ell \le 2j. \tag{5.4}$$

Then the integrated rates in (5.2) satisfy

$$\frac{r_k}{t} = \sum_{\ell=1}^{2j}(2\ell+1)(-1)^{2j+k+\ell+1}\begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix}\chi_\ell. \tag{5.5}$$

After tetrahedral symmetry, the inequalities $r_k \geq 0$ are the single-spin inequalities of [9, Eq. (2.20)]. For $j = 1$, they are

$$\frac{3}{5}\chi_1 \leq \chi_2 \leq 3\chi_1, \tag{5.6}$$

which is [9, Eq. (2.21)].

Proof. Substitute $\log\eta_\ell = -t\chi_\ell$ into (5.2); this gives (5.5). For $j = 1$, the two primitive rows $(3,-5)$ and $(-3,1)$ in logarithmic coordinates become $(-3,5)$ and $(3,-1)$ in decay-rate coordinates, which is equivalent to (5.6). ∎

symbol	role	defining relation
p_k	channel-simplex weight	$\Phi = \sum_k p_k\mathcal{E}_k$, $p_k \geq 0$, $\sum_k p_k = 1$
α_k	static channel weight	$p_k = (2k+1)\alpha_k$ [18, Eqs. (26)–(31)]
γ_k	generator rate	$\mathcal{L} = \sum_{k\geq 1}\gamma_k\mathcal{D}_k$, $\gamma_k \geq 0$
λ_ℓ	generator eigenvalue on V_ℓ	$\boldsymbol{\lambda} = M^{(j)}\boldsymbol{\gamma}$
η_ℓ	channel eigenvalue on V_ℓ	$\eta_\ell = e^{t\lambda_\ell}$ for a semigroup endpoint
r_k	integrated generator rate	$\boldsymbol{r} = (M^{(j)})^{-1}\log\boldsymbol{\eta} = t\boldsymbol{\gamma}$
χ_ℓ	historical sector decay rate	$\chi_\ell = -t^{-1}\log\eta_\ell$

Table 1. Weight, rate, and sector-eigenvalue conventions used in the paper.

Proposition 5.4 *(Poisson collision representation)***.** *For $r_k \geq 0$, put $a_k = (2k+1)r_k/d$. The semigroup endpoint with integrated rates $r = (r_1,\ldots,r_{2j})$ factors as*

$$\exp\left(\sum_{k=1}^{2j} r_k\mathcal{D}_k\right) = \prod_{k=1}^{2j}\left[e^{-a_k}\sum_{m=0}^{\infty}\frac{a_k^m}{m!}\mathcal{E}_k^m\right]. \tag{5.7}$$

Thus each extreme generator ray is the continuous-time Poissonization of repeated applications of one Clebsch–Gordan vertex channel, and a general covariant semigroup is the product of these commuting processes.

Proof. Sector diagonalization makes the maps $\mathcal{D}_k$ commute. Equation (2.16) gives

$$e^{r_k\mathcal{D}_k} = e^{a_k(\mathcal{E}_k - \mathrm{id})} = e^{-a_k}\sum_{m=0}^{\infty}\frac{a_k^m}{m!}\mathcal{E}_k^m.$$

The series converges in operator norm. Multiplying these identities and using commutativity proves (5.7). ∎

Corollary 5.5 *(Spin 1/2 baseline)***.** *For $j = 1/2$, the covariant-channel interval is $-1/3 \leq \eta_1 \leq 1$, while*

$$M^{(1/2)} = (-2), \qquad r_1 = -\frac{1}{2}\log\eta_1. \tag{5.8}$$

Consequently a spin-$1/2$ *covariant channel is covariantly embeddable if and only if* $0 < \eta_1 \leq 1$*. Thus every channel in this one-dimensional simplex with strictly positive sector spectrum is embeddable.*

Proof. The two channel vertices have eigenvalues 1 and $-1/3$, by (4.11). Equation (4.15) gives $M^{(1/2)} = -2$, and Theorem 5.1 gives (5.8). ■

Corollary 5.6 *(Spin* 1*)***.** *For* $j = 1$,

$$M^{(1)} = \begin{pmatrix} -\frac{1}{2} & -\frac{5}{2} \\ -\frac{3}{2} & -\frac{3}{2} \end{pmatrix}, \qquad (M^{(1)})^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{5}{6} \\ -\frac{1}{2} & \frac{1}{6} \end{pmatrix}. \tag{5.9}$$

Writing $x_\ell = \log \eta_\ell$*, a covariant channel is covariantly embeddable if and only if* $\eta_1, \eta_2 > 0$ *and*

$$3x_1 - 5x_2 \geq 0, \qquad -3x_1 + x_2 \geq 0. \tag{5.10}$$

Equivalently,

$$\eta_1^3 \geq \eta_2^5, \qquad \eta_2 \geq \eta_1^3. \tag{5.11}$$

Proof. Substitution of the four relevant $6j$ symbols into (4.15) gives (5.9). The two rows of its inverse, after multiplication by the positive denominator 6, give (5.10). Exponentiation gives (5.11). ■

Corollary 5.7 *(Spin* $3/2$*)***.** *For* $j = 3/2$,

$$M^{(3/2)} = \begin{pmatrix} -\frac{1}{5} & -1 & -\frac{14}{5} \\ -\frac{3}{5} & -2 & -\frac{7}{5} \\ -\frac{6}{5} & -1 & -\frac{9}{5} \end{pmatrix} \tag{5.12}$$

and

$$(M^{(3/2)})^{-1} = \frac{1}{20} \begin{pmatrix} 11 & 5 & -21 \\ 3 & -15 & 7 \\ -9 & 5 & -1 \end{pmatrix}. \tag{5.13}$$

Writing $x_\ell = \log \eta_\ell$*, a covariant channel is covariantly embeddable if and only if every* $\eta_\ell > 0$ *and*

$$11x_1 + 5x_2 - 21x_3 \geq 0, \tag{5.14}$$

$$3x_1 - 15x_2 + 7x_3 \geq 0, \tag{5.15}$$

$$-9x_1 + 5x_2 - x_3 \geq 0. \tag{5.16}$$

Equivalently,

$$\eta_1^{11}\eta_2^5 \geq \eta_3^{21}, \qquad \eta_1^3\eta_3^7 \geq \eta_2^{15}, \qquad \eta_2^5 \geq \eta_1^9\eta_3. \tag{5.17}$$

Proof. Substitution in (4.15) gives (5.12); exact matrix inversion gives (5.13). The rows of the inverse give the three inequalities, and exponentiation gives (5.17). ■

Corollary 5.8 *(Spin 2)*. *For $j = 2$,*

$$M^{(2)} = \begin{pmatrix} -\frac{1}{10} & -\frac{1}{2} & -\frac{7}{5} & -3 \\ -\frac{3}{10} & -\frac{17}{14} & -\frac{11}{5} & -\frac{9}{7} \\ -\frac{3}{5} & -\frac{11}{7} & -\frac{9}{10} & -\frac{27}{14} \\ -1 & -\frac{5}{7} & -\frac{3}{2} & -\frac{25}{14} \end{pmatrix} \tag{5.18}$$

and

$$(M^{(2)})^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & -\frac{6}{5} \\ \frac{3}{10} & -\frac{3}{14} & -\frac{4}{5} & \frac{18}{35} \\ 0 & -\frac{4}{7} & \frac{1}{2} & -\frac{9}{70} \\ -\frac{2}{5} & \frac{2}{7} & -\frac{1}{10} & \frac{1}{70} \end{pmatrix}. \tag{5.19}$$

Writing $x_\ell = \log \eta_\ell$, a covariant channel is covariantly embeddable if and only if every $\eta_\ell > 0$ and

$$5x_1 + 5x_2 - 12x_4 \geq 0, \tag{5.20}$$

$$21x_1 - 15x_2 - 56x_3 + 36x_4 \geq 0, \tag{5.21}$$

$$-40x_2 + 35x_3 - 9x_4 \geq 0, \tag{5.22}$$

$$-28x_1 + 20x_2 - 7x_3 + x_4 \geq 0. \tag{5.23}$$

Equivalently,

$$\eta_1^5 \eta_2^5 \geq \eta_4^{12}, \qquad \eta_1^{21} \eta_4^{36} \geq \eta_2^{15} \eta_3^{56}, \tag{5.24}$$

$$\eta_3^{35} \geq \eta_2^{40} \eta_4^{9}, \qquad \eta_2^{20} \eta_4 \geq \eta_1^{28} \eta_3^{7}. \tag{5.25}$$

Proof. Equations (4.15) and (4.17) give (5.18) and (5.19). Multiplying the four inverse rows by $10, 70, 70, 70$, respectively, and dividing by their positive common factors gives the primitive facet normals in (5.20)–(5.23). Exponentiation gives (5.24)– (5.25). ∎

6. Low-spin channel geometry

Proposition 6.1 *(Low-spin complete-positivity facets)*. *For spin 1, a real pair (η_1, η_2) is the sector spectrum of a covariant channel if and only if*

$$1 + 3\eta_1 + 5\eta_2 \geq 0, \tag{6.1}$$

$$2 + 3\eta_1 - 5\eta_2 \geq 0, \tag{6.2}$$

$$2 - 3\eta_1 + \eta_2 \geq 0. \tag{6.3}$$

For spin 3/2, the corresponding four inequalities are

$$1 + 3\eta_1 + 5\eta_2 + 7\eta_3 \geq 0, \tag{6.4}$$

$$5 + 11\eta_1 + 5\eta_2 - 21\eta_3 \geq 0, \tag{6.5}$$

$$5 + 3\eta_1 - 15\eta_2 + 7\eta_3 \geq 0, \tag{6.6}$$

$$5 - 9\eta_1 + 5\eta_2 - \eta_3 \geq 0. \tag{6.7}$$

Proof. For spin 1, the three barycentric coordinates (4.23) are

$$\frac{1+3\eta_1+5\eta_2}{9}, \qquad \frac{2+3\eta_1-5\eta_2}{6}, \qquad \frac{5(2-3\eta_1+\eta_2)}{18}.$$

For spin 3/2, they are

$$\frac{1+3\eta_1+5\eta_2+7\eta_3}{16}, \quad \frac{3(5+11\eta_1+5\eta_2-21\eta_3)}{80},$$
$$\frac{5+3\eta_1-15\eta_2+7\eta_3}{16}, \quad \frac{7(5-9\eta_1+5\eta_2-\eta_3)}{80}.$$

Corollary 4.7 proves the claim. ∎

Proposition 6.2 *(Exact comparison polytopes)***.** *In tensor-eigenvalue coordinates, the spin-*1 *covariant-channel simplex and its nonnegative-spectrum part are*

$$\mathsf{S}_1 \;=\; \operatorname{conv}\left\{(1,1), \left(\tfrac{1}{2}, -\tfrac{1}{2}\right), \left(-\tfrac{1}{2}, \tfrac{1}{10}\right)\right\}, \tag{6.8}$$
$$\mathsf{S}_1^+ \;=\; \operatorname{conv}\left\{(0,0), \left(0, \tfrac{2}{5}\right), \left(\tfrac{2}{3}, 0\right), (1,1)\right\}. \tag{6.9}$$

For spin 3/2, the full simplex is

$$\begin{aligned}\mathsf{S}_{3/2} \;=\; \operatorname{conv}\{&(1,1,1), \left(\tfrac{11}{15}, \tfrac{1}{5}, -\tfrac{3}{5}\right), \left(\tfrac{1}{5}, -\tfrac{3}{5}, \tfrac{1}{5}\right),\\ &\left(-\tfrac{3}{5}, \tfrac{1}{5}, -\tfrac{1}{35}\right)\},\end{aligned} \tag{6.10}$$

and its nonnegative-spectrum part has the eight vertices

$$\begin{gathered}(0,0,0), \quad \left(0,0,\tfrac{5}{21}\right), \quad \left(0,\tfrac{1}{3},0\right), \quad \left(0,\tfrac{1}{2},\tfrac{5}{14}\right),\\ \left(\tfrac{1}{2},0,\tfrac{1}{2}\right), \quad \left(\tfrac{5}{9},0,0\right), \quad \left(\tfrac{5}{6},\tfrac{1}{2},0\right), \quad (1,1,1).\end{gathered} \tag{6.11}$$

Proof. Substitution of $j = 1$ and $j = 3/2$ in (4.11) gives (6.8) and (6.10). For the nonnegative-spectrum intersections, write a point as $\eta = V\boldsymbol{p}$, where the columns of V are the displayed simplex vertices and $p_k \geq 0$, $\sum_k p_k = 1$. A vertex of the intersection with $\eta_\ell \geq 0$ is the unique solution of a full-rank selection of the bounding equalities $p_k = 0$ and $\eta_\ell = 0$. The exact enumeration in `code/generate_plot_data.py` solves every such rational system and tests all remaining inequalities. Its certificate `paper/generated/geometry_exact.txt` is (6.9) and (6.11); exhaustion of all full-rank active sets proves that there are no further vertices. ∎

Proposition 6.3 *(Euclidean volume of the embeddable region)***.** *Let $n = 2j$, let $\mathsf{M}_j \subset \mathbb{R}^n$ be the set of covariantly embeddable sector spectra, and define*

$$c_k \;=\; -\sum_{\ell=1}^{n} M_{\ell k}^{(j)} \;=\; 2k, \qquad 1 \,\leq\, k \,\leq\, n. \tag{6.12}$$

With respect to Lebesgue measure in the fixed sector-eigenvalue coordinates,

$$\operatorname{vol}_n(\mathsf{M}_j) \;=\; \frac{2j+1}{\prod_{k=1}^{n} c_k} \;=\; \frac{2j+1}{2^n n!}. \tag{6.13}$$

For spins 1, 3/2, *and* 2, *the column-decay vectors and volumes are*

$$\begin{array}{c|c|c} j & (c_1,\ldots,c_{2j}) & \mathrm{vol}_{2j}(\mathsf{M}_j) \\ \hline 1 & (2,4) & 3/8 \\ 3/2 & (2,4,6) & 1/12 \\ 2 & (2,4,6,8) & 5/384. \end{array} \tag{6.14}$$

Moreover,

$$\mathrm{area}(\mathsf{S}_1) \;=\; \frac{9}{10}, \qquad\qquad \mathrm{area}(\mathsf{S}_1^+) \;=\; \frac{8}{15}, \tag{6.15}$$

$$\mathrm{vol}_3(\mathsf{S}_{3/2}) \;=\; \frac{128}{315}, \qquad\qquad \mathrm{vol}_3(\mathsf{S}_{3/2}^+) \;=\; \frac{25}{126}. \tag{6.16}$$

Thus the embeddable fractions of the nonnegative-spectrum regions are 45/64 *for spin* 1 *and* 21/50 *for spin* 3/2*; the fractions of the full channel simplices are* 5/12 *and* 105/512*, respectively.*

Proof. Let $\Delta_j \subset B(\mathcal{H}_j)$ be the d-dimensional subspace of matrices diagonal in the angular-momentum basis. The selection rule in (2.3) is

$$\langle j,m|T^{(k)}_q|j,a\rangle \;\neq\; 0 \quad\Longrightarrow\quad m-a \;=\; q. \tag{6.17}$$

If $X \in \Delta_j$, a nonzero summand in $\langle j,m|T^{(k)}_q X (T^{(k)}_q)^\dagger|j,n\rangle$ requires both $m-a = q$ and $n-a = q$ for the same diagonal index a, and therefore $m = n$. Hence the Kraus map

$$S_k(X) \;=\; \sum_{q=-k}^{k} T^{(k)}_q X (T^{(k)}_q)^\dagger$$

preserves Δ_j. In the orthonormal diagonal matrix-unit basis, its restricted superoperator trace is

$$\mathrm{Tr}(S_k|_{\Delta_j}) \;=\; \sum_{m=-j}^{j}\sum_{q=-k}^{k} |\langle j,m|T^{(k)}_q|j,m\rangle|^2 \;=\; \sum_{m=-j}^{j} |\langle j,m|T^{(k)}_0|j,m\rangle|^2 \;=\; 1,$$

where the last equality is the Hilbert–Schmidt normalization of $T^{(k)}_0$. The negative scalar term in (2.8) contributes $-(2k+1)$ to the restricted trace, so

$$\mathrm{Tr}(\mathcal{D}_k|_{\Delta_j}) \;=\; -2k. \tag{6.18}$$

The operators $T^{(\ell)}_0$, $0 \le \ell \le n$, form an orthonormal eigenbasis of Δ_j. The eigenvalue for $\ell = 0$ is zero and the remaining eigenvalues are $M^{(j)}_{\ell k}$. Comparing their sum with (6.18) proves (6.12).

By Theorem 5.1, the map

$$F : \mathbb{R}^n_{\ge 0} \longrightarrow \mathsf{M}_j, \qquad F(r) \;=\; \exp(M^{(j)} r)$$

is one-to-one and onto, with componentwise exponential. In the interior its Jacobian is

$$DF(r) \;=\; \mathrm{diag}(F(r))M^{(j)}.$$

Equations (4.18) and (6.12) give

$$|\det DF(r)| \;=\; (2j+1)\exp\left(-\sum_{k=1}^{n} c_k r_k\right).$$

The boundary of the orthant is the finite union of its coordinate hyperplanes. On its intersection with every compact box, F is Lipschitz, so the image of each such $(n-1)$-dimensional piece has zero n-dimensional measure; a countable exhaustion by boxes shows that the entire boundary image is null. Thus the change-of-variables theorem on the interior loses no volume. Integrating this product over $\mathbb{R}^n_{\geq 0}$, and using $\prod_{k=1}^{n} 2k = 2^n n!$, proves (6.13).

Specializing (6.12) gives (6.14). The first entries in (6.15) and (6.16) are simplex determinants. The spin-1 nonnegative-spectrum area follows from the shoelace formula applied to (6.9). Triangulating each exact face in (6.11) from an interior point gives 25/126; the rational active-set script records the same determinant sum. Division gives the four stated fractions. ∎

Figure 1 and Figures 2–3 compare these polytopes with the exponential image of the generator cone; the zero-spectrum limit is shown only as part of its closure. The spin-1 boundary curves are exact. For spin 3/2, the rendered boundary meshes truncate the nonnegative rates at 8; the inequalities (5.14)–(5.16) remain the exact description.

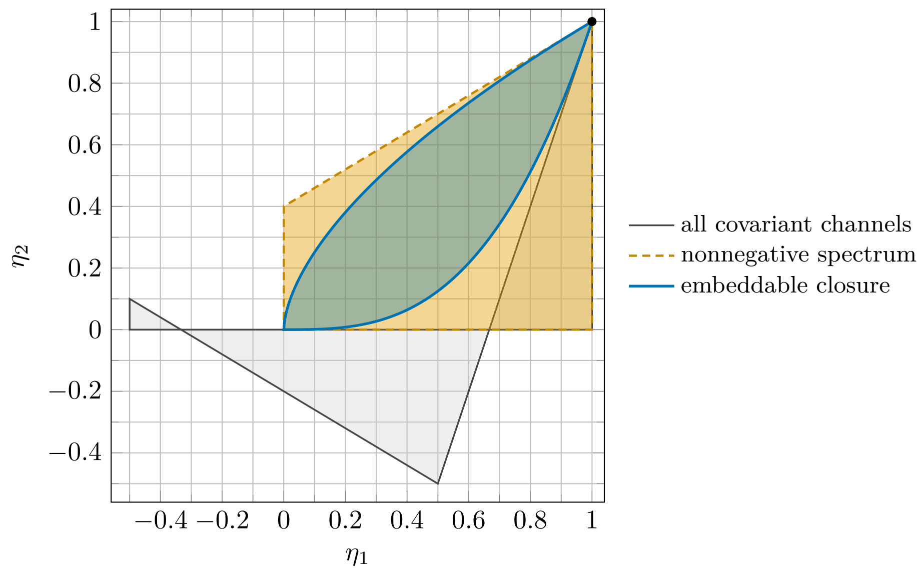
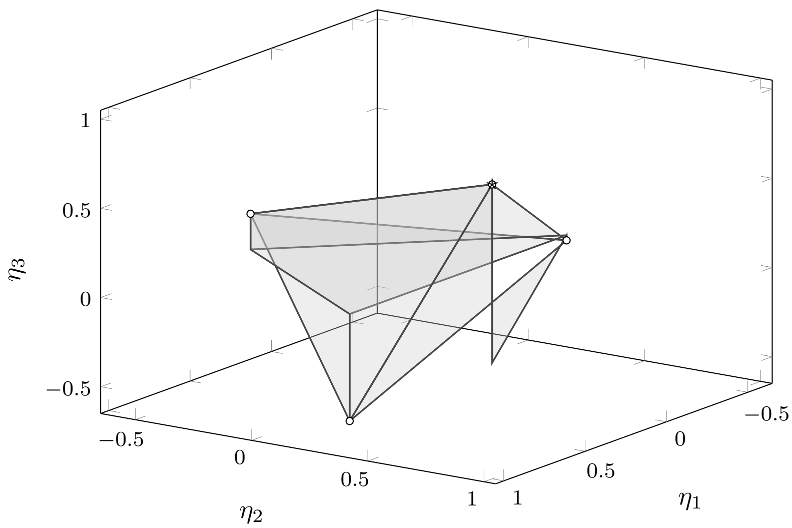
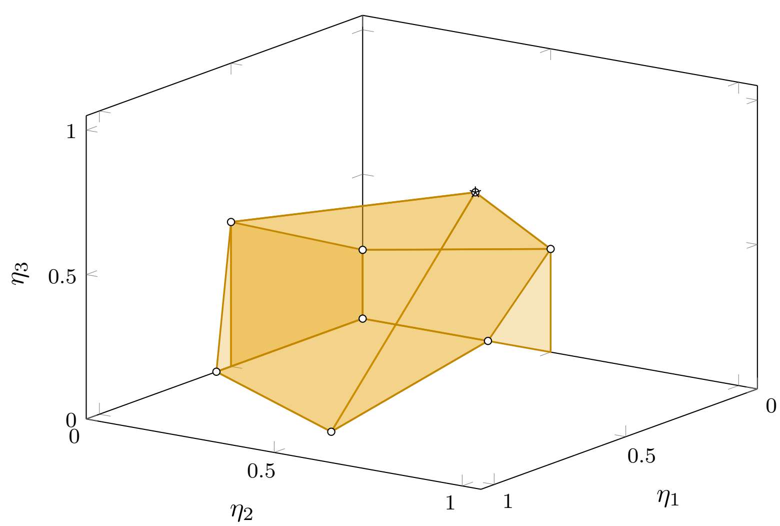


Figure 1. Spin 1. The gray triangle is the exact covariant-channel simplex. The orange polygon is its intersection with $\eta_1, \eta_2 \geq 0$. The blue region is the closure of the covariantly embeddable set and is bounded by $\eta_2 = \eta_1^3$ and $\eta_2 = \eta_1^{3/5}$, the two extreme rays of the logarithmic generator cone; $(0, 0)$ itself is not a finite-time endpoint.



(a) all channels



(b) nonnegative spectrum

Figure 2. Spin 3/2 static geometry. Both panels are generated from exact rational vertices: panel (a) is the full covariant-channel simplex, and panel (b) is its intersection with the nonnegative spectral orthant. Every vertex is marked; the identity vertex is starred.

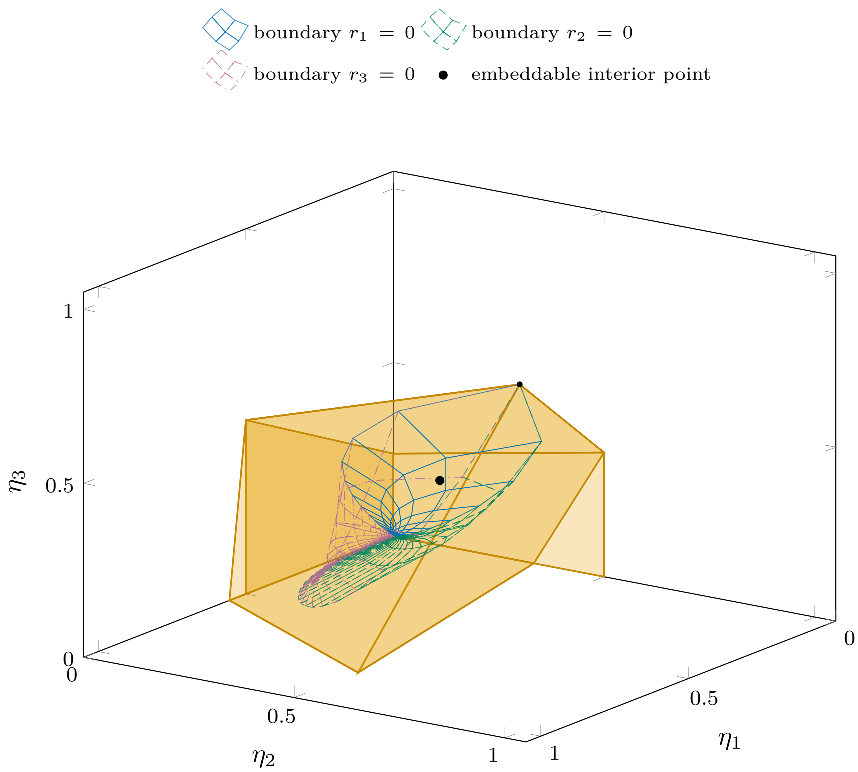


Figure 3. Spin 3/2 embeddable geometry. The three boundary meshes are overlaid on the orange nonnegative-spectrum polytope and are distinguished by color and line style; each label names the canonical rate set to zero. The embeddable region is the common side of the sheets containing the marked diagonal point $e^{-1}(1, 1, 1)$. The exact region is defined by inequalities (5.14)– (5.16). For rendering, the two free rates on each mesh range over $[0, 8]$; the gray dotted edges are sampling cutoffs, not mathematical boundaries.

7. Differentiable CP-divisible paths

Definition 7.1 (CP-divisible channel paths)**.** Let $I \subset \mathbb{R}$ be a nondegenerate interval. A family of channels $\{\Phi_t\}_{t\in I}$ is CP-divisible if there are channels $\mathcal{V}_{t,s}$ for every $s \le t$ in I such that

$$\Phi_t = \mathcal{V}_{t,s}\Phi_s, \qquad \mathcal{V}_{t,t} = \mathrm{id}, \qquad \mathcal{V}_{u,t}\mathcal{V}_{t,s} = \mathcal{V}_{u,s} \quad (s \le t \le u). \tag{7.1}$$

The channel-path definition does not itself require $\Phi_{t_0} = \mathrm{id}$ at any distinguished time. When this equality holds, we call the path identity-normalized at t_0. For the invertible paths considered below, the first identity in (7.1) uniquely determines the propagators, and the other two identities follow. Thus fixed negative sector signs can occur only under the broader, nonidentity-normalized convention. A C^1 path on I below has a continuous derivative in the interior, with the corresponding one-sided continuous derivative at each included endpoint. Norms of superoperators, including those in remainder and convergence statements, are the operator norms induced by the Hilbert–Schmidt norm.

The next lemma uses Choi's matrix criterion for complete positivity [5, Theorem 2] to describe the tangent cone at the identity, written in the present vectorization convention. We include the finite-dimensional derivation to make the normalization explicit.

Lemma 7.2 *(Compressed-Choi characterization)***.** *Let $K : B(\mathbb{C}^d) \to B(\mathbb{C}^d)$ be Hermiticity preserving and trace annihilating. With the unnormalized Choi convention*

$$J(K) = \sum_{a,b=1}^{d} K(|e_a\rangle\langle e_b|) \otimes |e_a\rangle\langle e_b|, \qquad \Omega = \sum_{a=1}^{d} e_a \otimes e_a, \tag{7.2}$$

put $Q = I - |\Omega\rangle\langle\Omega|/d$. Then K is a GKSL generator if and only if

$$QJ(K)Q \ge 0. \tag{7.3}$$

Proof. Choose a Hilbert–Schmidt orthonormal basis satisfying

$$F_0 = I/\sqrt{d}, \qquad \operatorname{Tr} F_a = 0 \quad (1 \le a \le d^2 - 1).$$

For $A \in B(\mathbb{C}^d)$, write $|A\rangle\rangle = (A \otimes I)\Omega$. Then $\{|F_a\rangle\rangle\}$ is orthonormal, $|F_0\rangle\rangle = |\Omega\rangle/\sqrt{d}$, and

$$J\big(X \mapsto AXB^\dagger\big) = |A\rangle\rangle\langle\langle B|. \tag{7.4}$$

Suppose first that (7.3) holds, and define

$$C_{ab} = \langle\langle F_a|J(K)|F_b\rangle\rangle, \qquad 1 \le a, b \le d^2 - 1.$$

Hermiticity preservation makes $J(K)$ Hermitian, and (7.3) says exactly that $C \ge 0$. Set

$$D_C(X) = \sum_{a,b=1}^{d^2-1} C_{ab}\left(F_a X F_b^\dagger - \frac{1}{2}\{F_b^\dagger F_a, X\}\right). \tag{7.5}$$

Equation (7.4) shows that

$$QJ(D_C)Q \;=\; \sum_{a,b=1}^{d^2-1} C_{ab}|F_a\rangle\!\rangle\langle\!\langle F_b| \;=\; QJ(K)Q;$$

the two anticommutator terms vanish under the compression because the Choi matrix of left or right multiplication has $|\Omega\rangle$ on one side.

Let $R = K - D_C$. Then $J(R)$ is Hermitian and $QJ(R)Q = 0$. Put $e_0 = \Omega/\sqrt{d}$. Relative to $\mathbb{C}e_0 \oplus \Omega^\perp$, there are $a \in \mathbb{R}$ and $z \in \Omega^\perp$ such that

$$J(R) \;=\; a|e_0\rangle\langle e_0| + |z\rangle\langle e_0| + |e_0\rangle\langle z|.$$

Choose $A \in B(\mathbb{C}^d)$ by

$$|A\rangle\!\rangle \;=\; \frac{a}{2\sqrt{d}}|e_0\rangle + \frac{1}{\sqrt{d}}|z\rangle.$$

Since $|I\rangle\!\rangle = \sqrt{d}\,|e_0\rangle$, this gives

$$J(R) \;=\; |A\rangle\!\rangle\langle\!\langle I| + |I\rangle\!\rangle\langle\!\langle A|.$$

By (7.4), $R(X) = AX + XA^\dagger$. Both K and D_C are trace annihilating, so

$$0 \;=\; \operatorname{Tr} R(X) \;=\; \operatorname{Tr}((A + A^\dagger)X) \qquad (X \in B(\mathbb{C}^d)).$$

Thus $A + A^\dagger = 0$. Writing $A = -iH$ with $H = H^\dagger$ gives $R(X) = -i[H, X]$, and (7.5) is a GKSL dissipator because $C \geq 0$.

Conversely, for a GKSL generator written in the same traceless basis, the Hamiltonian and anticommutator Choi terms vanish after compression, while the remaining compression is $\sum_{a,b} C_{ab}|F_a\rangle\!\rangle\langle\!\langle F_b| \geq 0$. This proves (7.3). ■

Lemma 7.3 *(Finite-dimensional local criterion)***.** *Let $t \mapsto \Phi_t$ be an invertible C^1 path of channels on a finite-dimensional matrix algebra. It is CP-divisible if and only if*

$$K_t \;=\; \dot{\Phi}_t\Phi_t^{-1} \tag{7.6}$$

is a GKSL generator for every t.

Proof. If the path is CP-divisible, invertibility makes the propagator unique: $\mathcal{V}_{t+h,t} = \Phi_{t+h}\Phi_t^{-1}$. It is a channel and

$$\mathcal{V}_{t+h,t} \;=\; \mathrm{id} + hK_t + o(h).$$

Let $\Omega = \sum_{a=1}^d e_a \otimes e_a$, use the unnormalized Choi convention $J(A) = \sum_{a,b} A(|e_a\rangle\langle e_b|) \otimes |e_a\rangle\langle e_b|$, and take $\psi \perp \Omega$. Since $J(\mathcal{V}_{t+h,t}) \geq 0$,

$$0 \;\leq\; \langle\psi, J(\mathcal{V}_{t+h,t})\psi\rangle \;=\; h\langle\psi, J(K_t)\psi\rangle + o(h). \tag{7.7}$$

Division by $h > 0$ and passage to the limit show that $J(K_t)$ is nonnegative on $\Omega^\perp$. Differentiating the trace-preserving and Hermiticity-preserving identities for $\mathcal{V}_{t+h,t}$ gives

$$\operatorname{Tr} K_t(X) \;=\; 0, \qquad K_t(X^\dagger) \;=\; K_t(X)^\dagger.$$

Thus K_t satisfies the hypotheses and compressed-Choi condition of Lemma 7.2, which gives its GKSL form. At a right endpoint of I, the same conclusion follows by continuity of K_t: the two linear identities above persist, and so does positivity of the compressed Choi matrix because the positive-semidefinite matrix cone is closed.

Conversely, suppose every K_t is a GKSL generator. Put $\Delta = (t-s)/n$. The chronological Euler products

$$\mathcal{V}_{t,s}^{(n)} = \exp(\Delta K_{s+(n-1)\Delta}) \cdots \exp(\Delta K_{s+\Delta}) \exp(\Delta K_s)$$

converge in operator norm to the solution of $\partial_t \mathcal{V}_{t,s} = K_t \mathcal{V}_{t,s}$, $\mathcal{V}_{s,s} = \mathrm{id}$. Indeed, if $B = \sup_{u \in [s,t]} \|K_u\|$ and

$$\omega_K(\delta) = \sup_{\substack{u,v \in [s,t] \\ |u-v| \le \delta}} \|K_u - K_v\|,$$

then variation of constants between the exact solution and the piecewise-constant evolution gives

$$\|\mathcal{V}_{t,s} - \mathcal{V}_{t,s}^{(n)}\| \le (t-s) e^{B(t-s)} \omega_K(\Delta) \longrightarrow 0.$$

Every factor is a channel, and the cone of completely positive maps and the trace-preserving affine subspace are operator-norm closed. Hence $\mathcal{V}_{t,s}$ is a channel. Finally, both Φ_t and $\mathcal{V}_{t,s}\Phi_s$ solve $\dot{Y}_t = K_t Y_t$ with initial value $Y_s = \Phi_s$; uniqueness for finite-dimensional linear ODEs gives $\Phi_t = \mathcal{V}_{t,s}\Phi_s$. ∎

Theorem 7.4 *(Instantaneous logarithmic cone)*. *Let $t \mapsto \Phi_t$ be an invertible C^1 path of* SU(2)*-covariant channels, with tensor-sector eigenvalues $\eta_\ell(t)$. Then the path is CP-divisible if and only if*

$$(M^{(j)})^{-1} \frac{d}{dt} \begin{pmatrix} \log|\eta_1(t)| \\ \vdots \\ \log|\eta_{2j}(t)| \end{pmatrix} \in \mathbb{R}^{2j}_{\ge 0} \quad \textit{for every } t. \tag{7.8}$$

If $\Phi_{t_0} = \mathrm{id}$ at some t_0, then every $\eta_\ell(t) > 0$ and the absolute values may be omitted.

Proof. Lemma 4.1 makes every $\eta_\ell(t)$ real, while invertibility makes it nonzero. Since the parameter set is an interval, continuity makes the sign of each η_ℓ constant. At an identity point all these signs are positive. Both $\dot{\Phi}_t$ and Φ_t^{-1} commute with every $\mathcal{U}_g$, so the time-local generator K_t in (7.6) is covariant. On V_ℓ it acts by

$$\frac{\dot{\eta}_\ell(t)}{\eta_\ell(t)} = \frac{d}{dt} \log|\eta_\ell(t)|.$$

Apply Lemma 7.3, then Theorems 3.1 and 4.5. ∎

Corollary 7.5 *(Spin-1/2 fixed-sign paths)*. *Let $t \mapsto \Phi_t$ be an invertible C^1 path of spin-1/2 covariant channels. Then it is CP-divisible if and only if*

$$\frac{d}{dt} \log|\eta_1(t)| \le 0.$$

Proof. Corollary 5.5 gives $(M^{(1/2)})^{-1} = -1/2$, so Theorem 7.4 reduces to the displayed inequality. ∎

Remark 7.6. Under Definition 7.1, every constant invertible channel path is CP-divisible, with identity propagators. In particular, this applies to the spin-1/2 nonidentity vertex $\eta_1 = -1/3$. The example illustrates the absence of identity normalization in the broader path convention; it does not embed that negative-spectrum vertex into a time-homogeneous quantum Markov semigroup issuing from the identity.

8. Exact computation and reproducibility

The symbolic calculation compares two algebraically distinct constructions: the Wigner $6j$ formula (4.15) and direct matrix operations on the tensors in (2.3). This separation makes the phase and normalization checks observable rather than folding them into a single table lookup; it is not a claim of independent software implementations.

Remark 8.1 (Exact reconstruction protocol). For $2j = 1, 2, 3, 4$, the repository's exact-cone script performs all of the following operations in symbolic arithmetic:

1. constructs every $T_q^{(k)}$ once from Clebsch–Gordan coefficients and once from the $3j$ formula (2.3), tests entrywise equality and the full Hilbert–Schmidt Gram matrix, and verifies $\sum_q (T_q^{(k)})^\dagger T_q^{(k)} = (2k+1)I/d$;
2. constructs the matrix of β in (4.4), tests its unitarity, and tests (4.7) for every k, q;
3. verifies $\mathcal{D}_k(I) = 0$, applies every Kraus sum directly to every $T_m^{(\ell)}$ with $1 \le \ell \le 2j$, and tests the full matrix identity $\mathcal{D}_k(T_m^{(\ell)}) = M_{\ell k}^{(j)} T_m^{(\ell)}$, and compares its scalar with (4.15);
4. constructs each $J(\mathcal{E}_k)$ separately from matrix units and from the tensor vectorizations, tests $J(\mathcal{E}_k) = dP_k/(2k+1)$, $P_k^2 = P_k$, $\operatorname{Tr} J(\mathcal{E}_k) = d$, and trace preservation;
5. tests (4.22), the closed inverse (4.17), $|\det M^{(j)}| = 2j+1$, and the general column identity $-\sum_\ell M_{\ell k}^{(j)} = 2k$.

The primitive integer rows printed by the script are exactly the facet normals in the spin-1, spin-3/2, and spin-2 corollaries. The tensor and Wigner routines return SymPy algebraic numbers [14], and the tests use exact matrix equality after symbolic simplification. The direct Kraus reconstruction does not call the $6j$-based generator routine. For every (ℓ, m, k), it checks the complete matrix residual and then compares the scalar with (4.15). The Choi matrix-unit loop is separate from its vectorized-projector construction. Thus items 1–4 compare distinct formula paths entry by entry; item 5 substitutes the resulting matrices into the displayed algebraic identities. The Clebsch–Gordan, $3j$, and $6j$ entry points all belong to SymPy's Wigner module, so shared library infrastructure is recorded rather than treated as an independent implementation. The executable check is

```
python3 -u code/exact_cones.py 1 2 3 4 -verify.
```

Remark 8.2 (Exact polytope certificates)*.* The script `code/generate_plot_data.py` enumerates every full-rank active set of the spin-1 and spin-3/2 simplex halfspaces together with $\eta_\ell \geq 0$, solves the systems over the rationals, and records each feasible vertex, its barycentric coordinates, the exact polytope volumes, and the embeddable-region Jacobian volumes in `paper/generated/geometry_exact.txt`. The only floating-point data produced by the script are plotting coordinates: the spin-1 curves and the truncated spin-3/2 boundary meshes. There are finitely many choices of as many active inequalities as the ambient dimension. A polytope vertex occurs in at least one full-rank choice. The script iterates over all choices, discards singular systems, solves the rest over $\mathbb{Q}$, and retains exactly the points that satisfy every halfspace. It then computes barycentric coordinates by the inverse augmented vertex matrix. Boundary cycles are ordered by exact signs of rational planar cross products. The certificate records those cycles and every rational determinant term in the shoelace and pyramid sums, so no floating-point decision enters a stated area or volume. This is the active-set argument used in Proposition 6.2.

From the repository root, `make check` runs the exact tensor audit, `make figures` regenerates all plot data and certificates, and `make paper` rebuilds the figures, bibliography, and paper. Both Python entry points accept `-h` and `-help` and exit without generating or changing artifacts.

9. Conclusion

For a fixed irreducible spin, the classical isotropic-spin generator–relaxation-rate transform takes a particularly transparent form in the present Hilbert–Schmidt normalization: covariance converts the canonical GKS matrix into one nonnegative rate per nonzero tensor rank, and the Racah transform converts those rates to sector decay exponents by $M^{(j)}$. Its orthogonality gives the inverse (4.17), while the additional determinant-magnitude identity and componentwise exponential turn this rate cone into an endpoint criterion. For identity-normalized paths, time-dependent CP-divisibility is the pointwise version of the same test applied to the logarithmic derivatives of the sector eigenvalues. For the broader channel-path convention, the signs are constant and the criterion uses the componentwise absolute values. The low-spin corollaries and comparison plots distinguish complete positivity, sectorwise nonnegativity, and covariant Markovian embeddability.

AI-use disclosure

OpenAI's GPT-5.6 Sol (`gpt-5.6-sol`), run at max reasoning effort, was used as an interactive research assistant during development of the proof strategy, stress testing of arguments, citation and calculation audits, preparation of exact verification code, and revision of the manuscript. The author set the research direction, assessed and corrected the resulting material, selected the final arguments, and assumes responsibility for every mathematical claim, citation, computation, and passage of text.

Code availability

The source, exact verification scripts, generated rational certificates, plotting data, and build instructions are available in the project repository. This reviewed manuscript revision is archived under the `v1.1.0` tag. Machine-readable citation metadata and the repository reuse terms are recorded in `CITATION.cff` and `LICENSE`, respectively. The saved exact-reconstruction transcript `paper/generated/exact_check.txt` has SHA-256 digest

`e95e71a454a726846654e2695eddc0a1ada95bdc4311fc5cbcc3b01a7e360ac9`.

The rational geometry certificate `paper/generated/geometry_exact.txt` has digest

`2c491ecb3cf638f6e9116cdeedd7970e38a02f0e84e1efc95429231e9178bad8`.

The repository records the package versions and the commands used to regenerate both certificates and the figures.

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